ENVIRONMENTAL HACKS 2026

AIR TRACK — PHASE 0 RESEARCH PAPER

AirGuard Decision Engine

Personalized, evidence-aware pollution exposure decision support

Project status: AIR TRACK LOCKED • Phase 0 restarted from zero • Coding NOT started

Research date: 8 October 2026 Team: Solo Track: Air

Research objective

Determine what is already available, what is already researched, what would be a weak copy, and what a solo developer can build during the hackathon that is differentiated, demonstrable, AWS-native and technically honest.

1. Problem Statement

Air pollution is not only a measurement problem. India already has extensive monitoring and public AQI communication. CPCB operates a continuous ambient air-quality monitoring network and publishes AQI information; its technical documentation describes hourly AQI publication and pollutant/station data infrastructure.

The remaining user problem is decision-making: “Given where I am going, when I am going, what I am doing, and the current pollution conditions, what should I change?” A single AQI number does not answer that question.

Examples of decisions include: whether to delay an outdoor activity, whether to choose a different route, whether a school should move activity indoors, whether an outdoor worker should shift the work window, and which evidence explains a pollution spike.

This project therefore treats AQI as an input—not the product.

Target user

Primary MVP user: urban commuter / student making a short outdoor journey. This narrow user keeps the product understandable and solo-feasible. Later extensions can support schools and outdoor workers.

Core question

“Which available route/time choice gives me lower estimated pollution exposure, what is the trade-off, and how confident is that recommendation?”

2. Hackathon Requirements and Strategy

The official Environmental Hacks page defines three tracks and explicitly places AQI, pollution exposure, stubble burning, indoor air and school safety under Air. It states that a small problem solved well beats a large problem solved vaguely. Judging includes Idea & Impact, Built on AWS, Design & Usability, Execution and the three-minute demo.

Rules state that project work starts when the hackathon opens; learning, planning and practice beforehand are allowed. Old projects do not count. The submission consists of a public repository, a YouTube demo of up to three minutes, and a short write-up. The demo must show AWS usage.

For this project, the consequence is straightforward: research and architecture can be prepared now; the actual repository implementation must begin within the hackathon window.

AWS strategy

Use AWS as a real system boundary rather than decoration. A minimal target is S3 for cached/raw data, Lambda for scoring, API Gateway for the API, EventBridge for refresh, DynamoDB for scenario/evidence records, CloudWatch for observability and Amplify/CloudFront for hosting. OpenAQ's AWS archive can also become a genuine AWS-native data source.

3. State of the Art / Existing Systems

A uniqueness claim must survive comparison with current systems and recent research. The following areas are already active:

3.1 AQI monitoring and public alerts

- CPCB already operates a real-time ambient monitoring network and publishes AQI and pollutant information. A dashboard that only visualizes these measurements adds little.
- CPCB technical plans also include city/station pollutant visualization, exceedance alarms, forecast integration and automatic severe/very-poor/poor AQI alerts. Therefore, alerting itself is not a sufficient novelty claim.

3.2 Cleaner / exposure-aware routing

- A 2026 India study proposed smart routing that integrates traffic congestion and air pollution exposure into a path-optimization model and reported simulated exposure reduction on a Mumbai corridor.
- A 2026 research paper on exposure-aware commuting uses Geo-AI for high-resolution NO2 modeling and route optimization.
- A September 2026 Journal of Transport & Health paper describes a personalized Healthy Route Planner for sensitive groups, using real-time air-quality mapping, user-specific health profiles and optimized routes.
- Commercial products such as CleanRoute already market cleaner-route and departure-time features. Therefore, “cleaner route” alone is not a unique project.

3.3 Personalized exposure

- Research platforms already combine location, ambient/community air-quality data, building/microenvironment information and activity to estimate personal exposure.
- Personalized exercise platforms already integrate air-quality data with exercise guidance.

3.4 Source attribution / pollution causes

- Current atmospheric research combines surface PM2.5 observations with chemical-transport models and emission inventories for source attribution. Therefore, causal source attribution is scientifically complex and should not be oversold in a hackathon MVP.
- Satellite active-fire data can provide useful evidence of fire anomalies, but fire detection is not proof that a particular nearby PM2.5 spike was caused by that fire.

4. Research Gap and Differentiation Thesis

The research does not support the claim that route optimization, personalized exposure, or AQI alerts are new. The defensible gap for this hackathon is the combination of decision context + evidence + uncertainty + trade-off + action in one compact user flow.

Proposed differentiation layers

1. Time × Route Decision

Compare a small number of candidate route/time scenarios instead of building a full navigation engine.

2. Pollutant-Specific Explanation

Identify the pollutant(s) driving the recommendation rather than treating AQI as a single opaque score.

3. Freshness-Aware Confidence

Show observation/model age and reduce confidence when data is stale or spatially distant.

4. Evidence Bundle

Every recommendation records source, timestamp, spatial resolution, key inputs, scoring version and confidence.

5. Explicit Trade-Off

State the cost of the safer option: extra minutes, distance or complexity. The user chooses knowingly.

6. Action Card

The result ends with a concrete recommendation rather than a graph: delay, choose route B, reduce outdoor intensity, or move activity indoors.

7. Human Context

The user can specify activity intensity/travel mode and time window; this changes the decision context without pretending to diagnose health.

8. Graceful Uncertainty

When data is missing or stale, the system says so and degrades instead of manufacturing a precise risk.

Novelty claim we can defend: AirGuard is not an invention of a new AQI algorithm. It is a transparent, evidence-aware decision layer that operationalizes existing air-quality data into a time/route/action recommendation with visible uncertainty and trade-offs.

5. Data Sources and Feasibility

Source	Role in MVP	Confidence / limitation
CPCB CAAQMS / AQI	Authoritative Indian observations and AQI	Station-based; not street-level everywhere
OpenAQ AWS archive	Machine-readable historical/global air-quality data	Provider licenses vary; coverage varies
IMD weather	Wind, temperature, humidity, forecast context	Meteorology does not directly equal roadside pollution
NASA FIRMS	Active-fire/thermal anomaly evidence	Evidence of fire activity, not causal PM2.5 proof
data.gov.in	Government supplementary datasets	Dataset/API availability varies
Route network	Candidate path/time alternatives	Need a permitted routing/geospatial source; do not assume a comm

OpenAQ is particularly attractive for AWS integration: its AWS Open Data Registry documents an hourly global archive, an S3 bucket and an API. AWS CLI access to the public archive does not require an AWS account. This gives us a credible data lineage story and a practical fallback dataset.

CPCB documentation confirms that India already has real-time monitoring infrastructure and hourly AQI publication. The product should therefore treat CPCB/OpenAQ as measurements to reason over, not as something we need to recreate.

Critical feasibility question: exact road-level pollution exposure cannot be inferred directly from one nearby station. The MVP should use relative exposure estimates across scenarios and clearly state assumptions.

6. Proposed Technical Model

The MVP does not need a black-box deep-learning model. A transparent scoring engine is more defensible within four days and easier to audit.

Conceptual pipeline

Air-quality observations + weather + route context + time window + user activity → feature normalization → route/time exposure score → confidence/freshness adjustment → action recommendation → evidence bundle

Candidate features

- PM2.5 / PM10 concentration where available
- NO2 / O3 / other available pollutant signals
- AQI / sub-index
- Observation age
- Distance from monitoring station / spatial coverage indicator
- Wind direction/speed and weather context where available
- Route length and estimated travel time
- Road/traffic proxy if available
- Departure time
- User activity mode / exposure duration

Relative exposure formulation (conceptual)

For each candidate scenario, compute a normalized exposure cost over the journey window. Apply a freshness and spatial-confidence modifier. Then compare scenarios rather than claiming absolute inhaled dose. The winning scenario is the lowest estimated exposure subject to a user-defined time/distance tolerance.

This is intentionally a decision model, not a medical model. The output should be 'lower estimated exposure' rather than 'safe/unsafe for your lungs'.

7. AWS Architecture

Component	AWS service	Purpose
Web app	Amplify Hosting / S3 + CloudFront	Public demo UI
API	API Gateway	Receives scenario requests
Compute	Lambda	Data normalization, scoring, recommendation
Raw/cache data	S3	Snapshots and historical files
Scenario/evidence store	DynamoDB	Recommendation/evidence records
Refresh	EventBridge	Scheduled data refresh
Observability	CloudWatch	Logs/errors/metrics
Optional explanation	Bedrock/Strands	Natural-language explanation only; deterministic score remains source of tru

The architecture is deliberately small. Do not add Kubernetes, SageMaker, OpenSearch or a large agent framework unless a concrete feature requires them. Service count is not technical depth.

A strong demo moment is the evidence panel: the judge can open a recommendation and see exactly which sources and timestamps produced it, plus the confidence and trade-off.

8. UX / Demo Design

The judges have no live call; the three-minute video is the product. Therefore the interface must communicate the value immediately.

Screen 1 — Decision input

Origin → destination → departure window → travel mode/activity → tolerance for extra time.

Screen 2 — Decision result

Recommended: Route B at 7:00 PM
Estimated relative exposure: 23% lower than fastest option
Travel cost: +6 minutes
Main pollutant driver: NO2
Confidence: 74%
Data freshness: 18 min

Screen 3 — Evidence

Station/source → timestamp → pollutant values → weather context → route comparison → confidence factors. This screen is the trust layer.

Screen 4 — What to do

One or two concrete actions. Avoid a dashboard wall of charts. The user should know what to change after one glance.

9. Solo MVP Scope and Timeline

Window	Deliverable	Definition of done
Oct 8	Repo + data adapter + UI skeleton + AWS skeleton	One public data source loads into the app
Oct 9	Scenario engine + route/time comparison	Two or more candidate scenarios produce a recommendati
Oct 10	AWS deployment + evidence + polish	Public URL works; AWS path visible
Oct 11	Testing + demo + submission	Clean-browser demo works under 3 minutes

Hard scope boundary: no turn-by-turn navigation, no custom air-quality sensors, no medical diagnosis, no large custom deep-learning training, no multi-city expansion until the core flow works.

Optional features only after MVP: school mode, outdoor-worker mode, fire-event context, push notifications, saved routes.

10. Failure Modes and Controls

- Existing product overlap: mitigate by focusing on evidence/freshness/trade-off/action rather than route cleanliness alone.
- Data gap in Lucknow: use the best-supported nearby monitoring sources and disclose spatial resolution; do not invent street-level values.
- Route API cost/limits: validate the route source before implementation; cache candidate routes.
- Stale air data: confidence penalty + visible timestamp + fallback state.
- Pollutant attribution overclaim: call it 'dominant signal' or 'associated evidence', not proven source causality.
- AI decoration: keep the scoring deterministic; use an LLM only to explain already-computed evidence.
- Demo overload: one user, one journey, two/three alternatives, one recommendation.

Red-line claims we will not make

- “This app predicts exactly what you will inhale.”
- “This route is medically safe.”
- “We are the first clean-air route planner.”
- “Satellite fire detection proves this local pollution spike.”
- “Our model gives street-level truth when the data is station/grid based.”

11. Phase 0 Research Questions to Close Before Coding

Phase 0 is restarted, but it is not complete yet. These questions are the final gates.

- Which CPCB/OpenAQ stations provide the most useful coverage for the selected demo geography?
- Which exact air-quality fields and timestamps can be retrieved reliably during the hackathon?
- Which weather endpoint can supply wind and timing context without creating unnecessary dependency risk?
- Which route/geospatial source can be legally and reliably used for the demo?
- Can relative route exposure be computed honestly from available spatial data?
- What is the minimum AWS architecture that can be deployed and shown in the video?
- Which single user persona gives the clearest problem-impact story?
- What exact metric will demonstrate value: relative exposure reduction, time/exposure trade-off, or decision confidence?
- What happens when no fresh station data exists?
- Can the entire flow be reproduced from a clean browser in under three minutes?

Phase 0 completion condition: these questions must have evidence-backed answers, the problem statement must be frozen, the data contract must be defined, and the MVP must be bounded before implementation begins.

12. Research Conclusion

AIR TRACK IS LOCKED. The project direction is AirGuard Decision Engine. The hackathon problem is not 'measure pollution' but 'help a person make a lower-exposure choice using transparent evidence'.

The research shows that AQI monitoring, alerts, cleaner routes, personalized exposure and health-aware routing already exist in various forms. Therefore, the project will differentiate through the combination and presentation of decision evidence: time × route × pollutant × freshness × confidence × trade-off × action.

The project should remain a small, reliable decision-support MVP. A transparent relative-exposure model is preferable to an impressive but unsupported ML claim. AWS should be visible in the actual data/compute/deployment path, not merely named in documentation.

The strongest final product story is:

“AirGuard tells you not just how bad the air is, but which available choice is better for your planned journey—and shows the evidence, uncertainty and trade-off behind that recommendation.”

Next phase: continue Phase 0 with exact API/data validation and competitor feature mapping. Only after those gates are satisfied will Phase 0 be marked COMPLETE and implementation start.

13. Source Register

Official hackathon

- WeMakeDevs Environmental Hacks overview: https://www.wemakedevs.org/aws/env
- WeMakeDevs Environmental Hacks rules: https://www.wemakedevs.org/aws/env/rules

Air data

- OpenAQ — Registry of Open Data on AWS: https://registry.opendata.aws/openaq/
- CPCB technical activity / CAAQMS and AQI: https://cpcb.nic.in/uploads/it_technical_activity.pdf
- CPCB National AQI: https://cpcb.nic.in/National-Air-Quality-Index.php
- NASA FIRMS: https://firms.modaps.eosdis.nasa.gov/
- India Open Government Data: https://data.gov.in/

Research / existing solutions

- 2026 India smart-route air pollution optimization: https://doi.org/10.3389/frsc.2026.1759665
- 2026 exposure-aware Geo-AI commuting: https://doi.org/10.1016/j.scs.2026.107403
- 2026 personalized Healthy Route Planner: https://doi.org/10.1016/j.jth.2026.102411
- TracMyAir personalized exposure platform: https://pmc.ncbi.nlm.nih.gov/articles/PMC12934869/
- P-STEP personalized exercise + air-quality platform: https://pmc.ncbi.nlm.nih.gov/articles/PMC12534489/
- CleanRoute: https://play.google.com/store/apps/details?id=com.auralabsstudio.cleanroute
- 2026 India PM2.5 source-attribution research: https://acp.copernicus.org/articles/26/10533/2026/acp-26-10533-2026.html
- India personal PM2.5 exposure research: https://www.nature.com/articles/s41370-019-0150-5

Research note: Reviewed 8 October 2026. “Unique” means a differentiated hackathon product concept based on the reviewed landscape; it is not a patent or legal novelty opinion.